

IMPEx Service Assessment and Contributions from the Graz Team

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IMPE_x SimDM

Comments and Recommendations



Acknowledgements

```
<SimulationModel>
```

```
  <ResourceID>
```

```
    impex://SINP/SimulationModel/Earth/Static
```

```
  </ResourceID>
```

```
  <ResourceHeader>
```

```
    <ResourceName>MF-PARABOLOID_SINP</ResourceName>
```

```
    <ReleaseDate>2012-09-21T00:00:00.000</ReleaseDate>
```

```
    <Description>...</Description>
```

```
    <Acknowledgement>Acknowledgement</Acknowledgement>
```

```
    <Contact>
```

```
      <PersonID>SINP</PersonID>
```

```
      <Role>DataProducer</Role>
```

```
    </Contact>
```

```
    <InformationURL>
```

```
      <URL>...</URL>
```

```
    </InformationURL>
```

```
  </ResourceHeader>
```

Comments (1)

- ◆ **ResourceId** naming rules in SPASE
 - ◆ See documentation:
<http://www.spase-group.org/school/tutorials/resourceid/>
 - ◆ SimDMs must conform to it (makes life easier for devs)
 - ◆ SINP already is conform (to some extend...)

Example:

- ◆ Model: `impex://SINP/SimulationModel/Earth/Static`
- ◆ Run: `impex://SINP/SimulationRun/Earth/2012-03-10UT12`

Agreement on construction of the path part (e.g. ../Earth/..)?

Comments (2)

Why naming rules are important?

Dynamic construction of a tree representation is easily enabled through **correct paths** encoded in the ResourceId

Example: How to get all simulation runs of a model with id
`impex://SINP/SimulationModel/Earth/Static?`

Search for elements with **id matching path**:
`impex://SINP/SimulationRun/Earth/Static/*`

Note: There is no clear restriction on the path element, but it must be logical (and context based)

Comments (3)

Alternative Association of Elements

- ◆ Granule → **ParentID** (NumericalOutput)
- ◆ NumericalOutput → **InputResourceID** (SimulationRun)
- ◆ SimulationRun → **Model.ModelID** (SimulationModel)
- ◆ SimulationModel → match the encoded repository

Comments (4)

- ◆ **Multiple files?** One per Repository?
 - ◆ FMI uses two Repositories with the **same** ResourceID (very bad)
- ◆ **File naming rule:** <authority_name>_tree.xml
 - ◆ Multiple files?
<authority_name>_<repository_name>_tree.xml
 - ◆ See current situation: <http://impex-portal.oeaw.ac.at/>
- ◆ IMPEX portal registry **consolidates** all trees:
 - ◆ <http://impex-portal.oeaw.ac.at/registry>
 - ◆ **See live examples later...**

Comments (5)

- ◆ Production versions of trees must **always validate** against an official version of the Schema.
 - ◆ Every **official release** is available from:
<http://impex-fp7.oeaw.ac.at/xsd/>
 - ◆ Documentation is available for all official versions (PDF + HTML)
- ◆ IMPEX portal always uses the **latest version**
 - ◆ Trees not validating against the latest version are **excluded**
- ◆ IMPEX portal **caches** the trees linked in the config for **24h**

Update Process

- ◆ Communication over a **separate** mailing list?
 - ◆ **Subject: [IMPE~~x~~ DM] ...**
 - ◆ Who is part of the SimDM consortium **after** IMPE~~x~~?
- ◆ Distribution of **development versions** of the tree?
 - ◆ With postfix “-dev” in file name?
- ◆ **Michel** is responsible for handling a **development version** of the SimDM Schema
 - ◆ Requests are **openly discussed** within the SimDM consortium
- ◆ Official release by **Michel and Florian** (through IMPE~~x~~ website)
 - ◆ Update of production versions with an announcement to the consortium (and tool maintainers)

IMPE_x Protocol

Comments and Recommendations



Comments (1)

- ◆ Please check and update **Documentation + WSDLs**
 - ◆ Methods definitions must be **inline** with ICDs and WSDLs
 - ◆ Some types are **not correct** in the spreadsheet
 - ◆ Spacecraft names are **not correct** at FMI (see ICD + WSDL)
- ◆ Changing WSDLs without notice is **breaking the service contract** (only technical 😊)
- ◆ JAVA libs don't like drastic SOAP interface changes
 - ◆ **Portal developments are on hold at the moment**

Comments (2)

- ◆ Please check the comments in:
<https://docs.google.com/document/d/1MMV00t472C0h91eKMdQgZCtOcXeOCYFCXbKBK63gQZI/edit#>
- ◆ **Are they still actual?**
 - ◆ e.g. *Why is the target in SINP methods not encoded as parameter?*
 - ◆ e.g. *Why do unsupported parameters not simply respond with SOAP faults?*
 - ◆ e.g. *Why do we import complicated types (resolving to xs:string) from the IMPEX DM? (see Taverna interoperability)*
- ... and also check the **Taverna assessment files** (testing matrix, modified WSDLs and link to myExperiment group)

Methods Definition (1)

Synchronisation of documentation in methods definition file (MD), WSDL & ICDs

- ◆ `getDataPointValue()` (SINP):
 - ◆ **Variable** and **InterpolationMethod** in WSDL but not in ICD
- ◆ `calculateDataPointValue()` (SINP)
 - ◆ extraParam **CoordinateSystem** in WSDL but not in ICD and MD
- ◆ `calculateCubeMercury()` (SINP):
 - ◆ **OutputFileType** in WSDLs but not in ICD and MD

Methods Definition (2)

- ◆ `getFileName()` (LATMOS):
 - ◆ **RunID** in ICD & MD but **ResourceID** in WSDL
- ◆ `getFieldLine()` (LATMOS):
 - ◆ extraParam **IMFClockAngle** in MD but not in ICD and WSDL
- ◆ `getParticleTrajectory()` (FMI):
 - ◆ **url_XYZ** in ICD and WSDL but **urlInitialParams** in MD
- ◆ `calculateDataPointValueFixedTime()` (SINP):
 - ◆ parameters **BD, BT, R1, R2, RD1 & RD2** in WSDL but not in ICD and MD

getMostRelevantRun Issues

- ◆ **Incorrect information in ICD:**

- ◆ `<scale>` parameter for `SW_function` is mandatory

- ◆ **Otherwise Error:**

```
<SOAP-ENV:Body>  
  <parameters>  
    error: Scale value cannot be zero: function  
  </parameters>  
</SOAP-ENV:Body>
```

- ◆ **General WS Problem at FMI: Error handling**

- ◆ Please use **SOAP-fault messages** (see later ...) or **return e.g.**

```
<parameters>  
  <error>Scale value cannot be zero: function</error>  
</parameters>
```

- ◆ Demo gives error too:

http://impex-fp7.fmi.fi/ws_tests/test_getMostRelevantRun.php

Why? Clients don't like literals with error info (see SINP and LATMOS)

Comments (3)

Indications on Portal developments

- ◆ REST API will include **all available** methods with metadata and testing UI (the new scoreboard)
 - ◆ Metadata is provided in **JSON format** (e.g. parameters and usage information)
 - ◆ **See examples in IMPEX infrastructure monitoring ...**
- ◆ Which methods do we want to have in the Portal UI (the IMPEX map) available?
 - ◆ **See proposals for use cases later ...**

TAVERNA interoperability

Why bother at all?

Extension of the accessibility **beyond** the scope of IMPEX

- ◆ Extremely versatile tool for **interconnection** of WS
- ◆ Taverna has now a special version with **SAMP** capabilities and speaks **IVOA protocols**

... but Taverna **doesn't like** the IMPEX WSDLs

→ Portal provides WSDLs which are working (see later)

TAVERNA interoperability

Report on current issues

- ◆ WSDL Schema imports **don't work**
 - ◆ Maybe an issue with namespaces?
- ◆ Taverna has some documentation on **WS compatibility**
 - ◆ See:
<http://www.taverna.org.uk/developers/web-service-developers/>
 - ◆ Taverna uses **WS-I specifications** (maybe there are restrictions there?)
 - ◆ Please (try to) use **SOAP fault messages** if incorrect parameters are sent ...

EPN-TAP interoperability

Yet another issue to care about ...

- ◆ EPN-TAP can be used with **TOPCAT**
- ◆ IVOA **Registry** entry for IMPEx resources is currently prepared
- ◆ **Preliminary use cases?**
 - ◆ **Granules** can definitely be displayed
 - ◆ See **D2.12_2**:
<https://docs.google.com/file/d/0B1erg7K1DCUqMnFxd1JWQkphVWc/edit>
 - ◆ Please try the service: http://epn-tap.ocaw.ac.at/_system_/tap/run/tap

... also check the translation documentation in **D2.12_2** and the used translation scripts at <https://github.com/FlorianTopf/impex-tools>

IMPEx infrastructure monitoring

General Error handling with WS – SOAP fault messages

- See W3C specs: <http://www.w3.org/TR/soap12-part1/#soapfault>

- Example: http://www.tutorialspoint.com/soap/soap_fault.htm

- SOAP-Body:

```
<SOAP-ENV:Fault>  
  <faultcode xsi:type="xsd:string">  
    SOAP-ENV:Client  
  </faultcode>  
  <faultstring xsi:type="xsd:string">  
    Scale value cannot be zero: function  
  </faultstring>  
</SOAP:ENV-Fault>
```

- Reason:** SOAP clients are handling faults natively and are returning according exceptions in their syntax. → **Makes the integration more reliable**

IMPEX infrastructure monitoring

Proposal: Monitoring through the IMPEX portal

- ◆ All service consumers **must use** the configuration:
<http://impex-portal.oeaw.ac.at/config>

Why?

- ◆ The IMPEX portal **monitors** all included resources (24h update cycle, immediate validation before usage)

The IMPEX portal also provides a **fully dynamic view** of its API: <http://impex-portal.oeaw.ac.at/api-view>

IMPEx infrastructure monitoring

Why bothering about the portal's API?

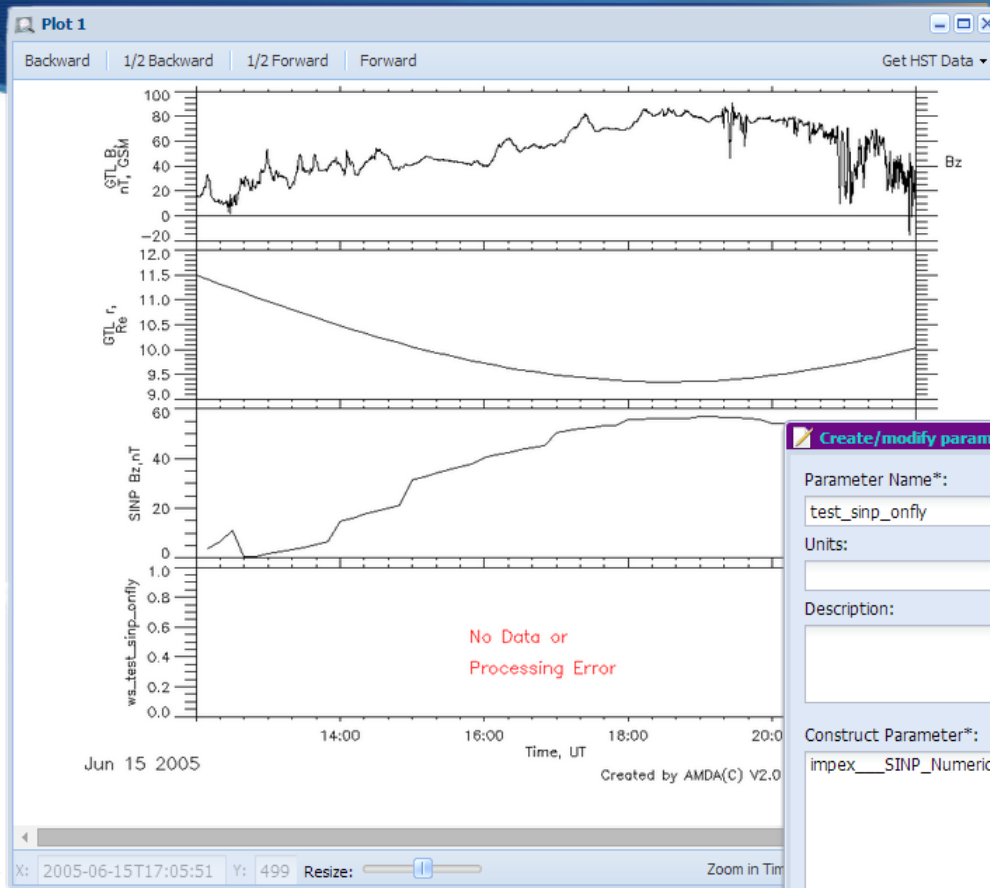
- ◆ The API is **REST-based** and gives access to
 - ◆ the IMPEx **configuration** (XML and JSON format),
 - ◆ a full IMPEx **registry** of selectable resources (XML and JSON format fully SPASE compliant)
 - ◆ a REST-based interface to all **IMPEx WS** (and WSDL files)
- ◆ The API is described by **SWAGGER**
 - ◆ See: <https://helloverb.com/developers/swagger>
 - ◆ Provides **JSON based metadata** for usage info:
<http://impex-portal.oeaw.ac.at/api-docs>

... and this info is used by the **API view**, try the result ☺

Current AMDA/3DView issues

- ◆ ~~SINP Earth on fly in AMDA:~~
 - ◆ ~~A2000 Earth on the fly is working (but only for one magnetic field component at once).~~
 - ◆ ~~Create/modify parameter is not working: returns error message “**No Data or Processing Error**”~~
 - ◆ ~~This functionality will be needed for the SINP Model Demonstrator~~

Current AMDA/3DView issues



Create/modify parameters

Parameter Name*: test_sinp_onfly

Time Step (sec)*: 60

Units:

Y Title for Plot:

Description:

Construct Parameter*:

impex___SNP_NumericalOutput_Earth_OnFly_Bx_By_Bz_Bz_Geotail

Save Reset

Information

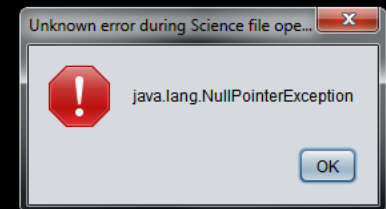
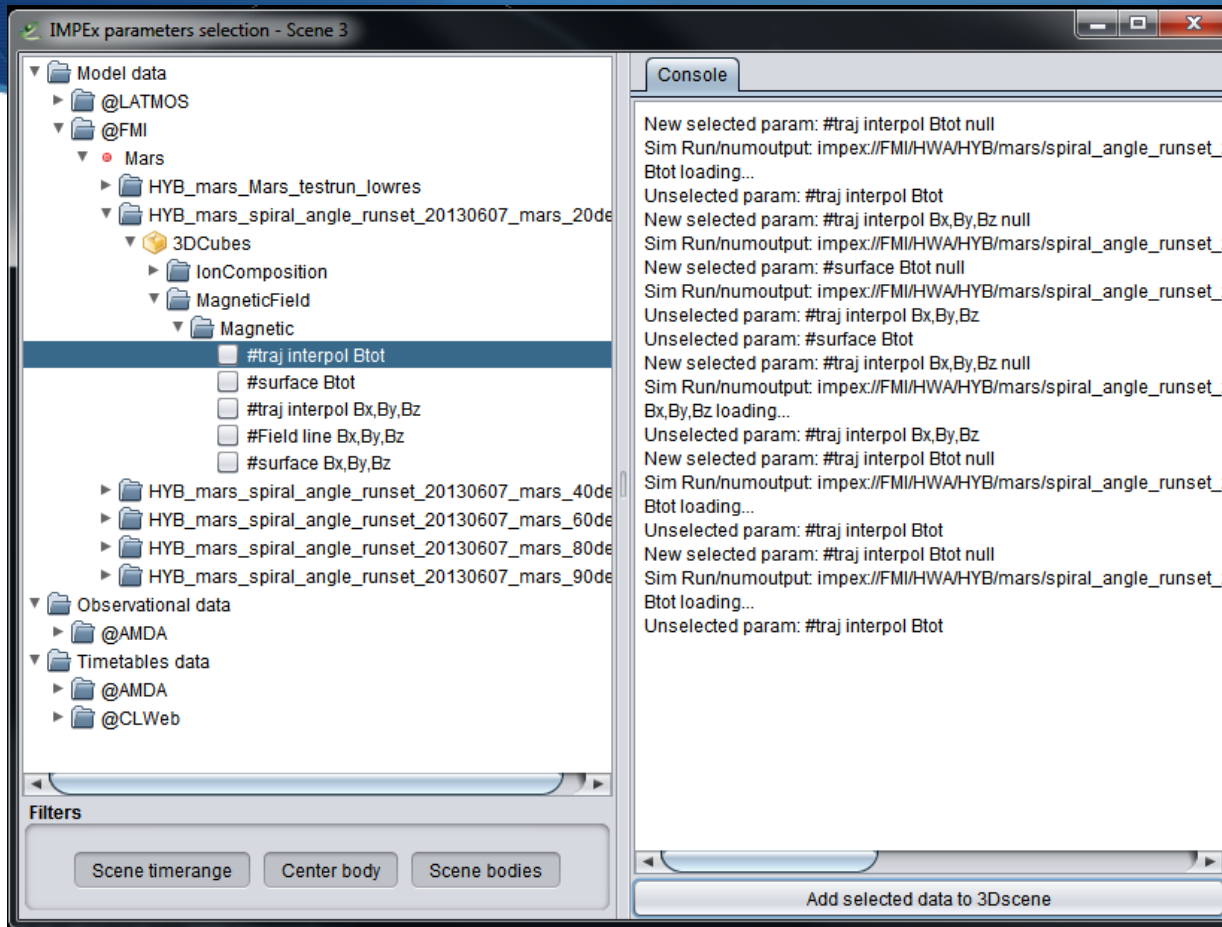
To construct a new parameter **drag** one of the AMDA parameters from the Parameters Tree and **drop** it onto the Construct Parameter text area.

Attention! Use only **LOW CASE** characters for parameter name

Current AMDA/3DView issues

- ❖ ~~FMI 3DCubes in 3DView:~~
 - ❖ ~~Total values of #traj interpol returns error message “java.lang.NullPointerException” for all cases~~

Current AMDA/3DView issues



Time: 2010-05-02T15:12:40

Distances (R_M = Mars radius = 3389,52666667km)

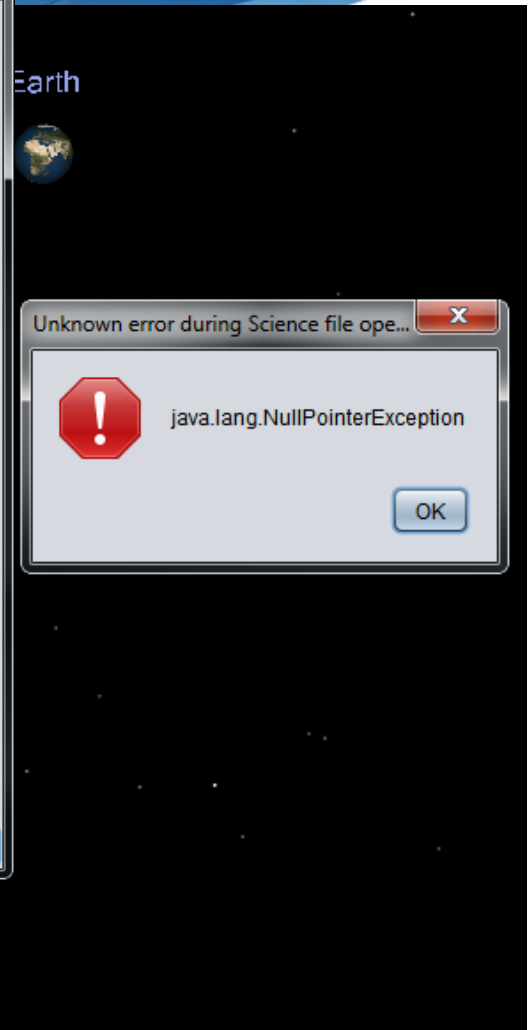
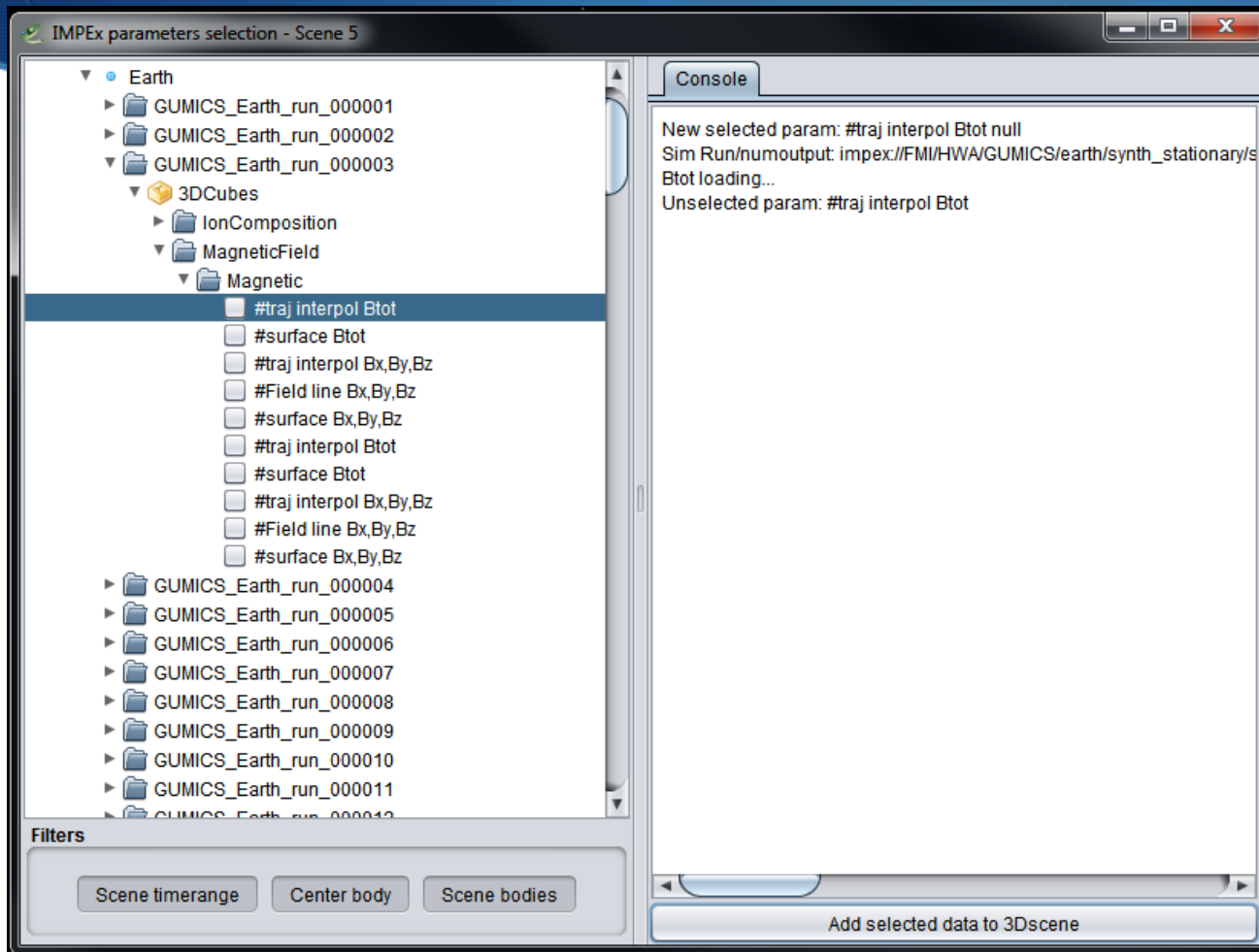
Frame = MSO

Center = Mars

Start = 2010/05/02 09:00:00

Stop = 2010/05/03 09:00:00

Current AMDA/3DView issues



time: 2012-05-02 10:00:00 Distances (R_E = Earth radius = 6371,010000000km)
Frame = GSE
Center = Earth
Start = 2012/05/02 10:00:00
Stop = 2012/05/03 10:00:00

IMPEx Portal

News and current Issues



Current Issues (1)

- ◆ **Access to Observational Data from AMDA**

- ◆ Problem with proprietary DM (**no official Schema**)

- ◆ Only **local tree** is used

- ◆ please never delete the files, just update them

- ◆ **Remote tree not necessary?**

- ◆ Some elements are excluded (e.g. models)

- ◆ **Translation to SPASE based tree?**

- ◆ Try the attempts e.g.

- ◆ <http://impex-portal.oeaw.ac.at/registry/observatory>

- ◆ Used Elements: Observatory, Instrument and NumericalData

... it really does make sense.

Current Issues (2)

Issues with AMDA/CLWeb to SPASE translation

Following information is **not available**:

- ◆ Observatory\ObservatoryRegion
- ◆ Instrument\InstrumentType
- ◆ NumericalData\MeasurementType
- ◆ NumericalData\ObservedRegion
- ◆ Parameter\Field\Quantity

Current translation table:

https://docs.google.com/spreadsheet/ccc?key=0A1erg7K1DCUqdHFCcjZROzR2UjZOM3lHeXpDYXNPQmc&usp=drive_web

Current Issues (3)

- ◆ **Resource filters in the map:**
 - ◆ **Target** (ObservatoryRegion or SimulatedRegion)
 - ◆ **Mission** (ResourceName)
 - ◆ I need a full mission list for IMPEx use cases!
 - ◆ Model data matching with resource id is possible
 - ◆ **InputField** (Bx, By, Bz), **StartTime**, **StopTime**?
- ◆ **Tree representation and browsing:**
 - ◆ See first attempt: <http://impex-portal.oeaw.ac.at/> (dynamic loading of tree selections via API)
 - ◆ Please note the issues with dynamic hierarchy...

Priorities

Current focus of developments

- ◆ Finalisation of **registry** (observational data?)
- ◆ Creation of **client framework** accessing the REST API
- ◆ **UI** representation of **databases** (simulation data only)
- ◆ **UI** representation of **REST API**
- ◆ Creation of **backend** enabling **WS access** (WSDLs should not change anymore)

Portal Messaging API

Please see documentation below:

- ◆ Contains all paths for **registry** and **config** access
- ◆ Contains paths to API **documentation**
- ◆ Contains preliminary selection of **data access** methods

◆ **API definition:**

https://docs.google.com/spreadsheets/ccc?key=0A1erg7K1DCUqdHVXZ2xjLVMzQ0pEN0RtOHcyVjhqS0E&usp=drive_web

◆ **Portal Sources:** <https://github.com/FlorianTopf/impex-portal>

Portal Messaging API

Registry Methods

Basic REST path:

GET /registry/{elementName}

- ◆ Parameters are simple **GET queries** (id and r for recursive)
- ◆ Documentation by **SWAGGER**
 - ◆ GET/api-docs/registry
- ◆ Note that the **element name** is used in the path
- ◆ **Filters** are available through path /registry/filter
- ◆ Result is always a valid **SPASE XML** document (or eq. JSON object)

Portal Messaging API

Data Access Methods

Basic REST path: (proposed)

GET /methods/{dbName}/{methodName}

- ◆ Method parameters are simple **GET queries**
- ◆ Documentation by **SWAGGER**
 - ◆ GET/api-docs/methods
- ◆ Note that the provider name part of the id is used in the path
- ◆ Additional paths to **WSDLs (proposed)**
 - ◆ GET /methods/{dbName} and GET /methods/taverna/{dbName}

Any complex issues I may have missed here?

New Portal Use Cases

- ◆ Which **data access methods** to we provide to the user?
- ◆ **What we want to have in the UI:**
 - ◆ All methods that can **produce VOTable** (+getMostRelevantRun())
 - ◆ All that can be used with **SAMP** for delegation
 - ◆ **Upload** possibility for **VOTable** files (other formats? ASCII?)
 - ◆ Example of **combined data usage**?
- ◆ Results are stored in a session-based user directory in VOTable format only. → **CRUD interface will be part of the API**

Notes:

https://docs.google.com/document/d/11WK1UOMc7p_PKmkrD0KesxesHvK1vOUNSmoiwNpJGTg/edit

Implementation Plan

- 💧 **Databases** (Trees view and selections)
- 💧 **Services** (WS views and access)
- 💧 **Data store** (VOTable management)
- 💧 **Filters** (Views and backend)
- 💧 **Extensions** for new **use cases** (TBD)
- 💧 **SAMP** access (Tool interoperability)

See Gantt chart:

<https://www.smartapp.com/gantterforgoogledrive/index.html?fileID=0B5hOkAv922Z6QUN1eEJISTZjQWs>

Conclusions

“If we could turn back the time, we would start with the portal...”

Yes and no...

... but the portal is a step towards a **service bus based solution** which was requested from the beginning.

Now we have it, spread the word!

The **portal client** should illustrate the **flexibilities** with this approach